Is the Web App Your AI Built Really Secure?

Find bugs.
Prove correctness.

01 / AI-LED RED-TEAMING

02 / FORMAL VERIFICATION

02 / THE FOUNDING INSIGHT

Software security has two sides

A lesson from my PhD: we need both.

01 / RED-TEAMING

Find Bugs ∃

Uncover whether vulnerabilities exist.

02 / FORMAL VERIFICATION

Prove Correctness ∀

Prove the security rules hold for all inputs.

h5i solves both sides of web security

01 / AI-LED RED-TEAMING

CLI-based Red-teaming

- Browsing
- Traffic capture
- Recon
- CI/CD

02 / FORMAL VERIFICATION

A Rust-based Web Framework

Rust → Lean 4

the agent-native workspace for web security

- Intuitive dashboard for humans
- Sandbox isolation for dangerous operations

AI operates. You oversee.

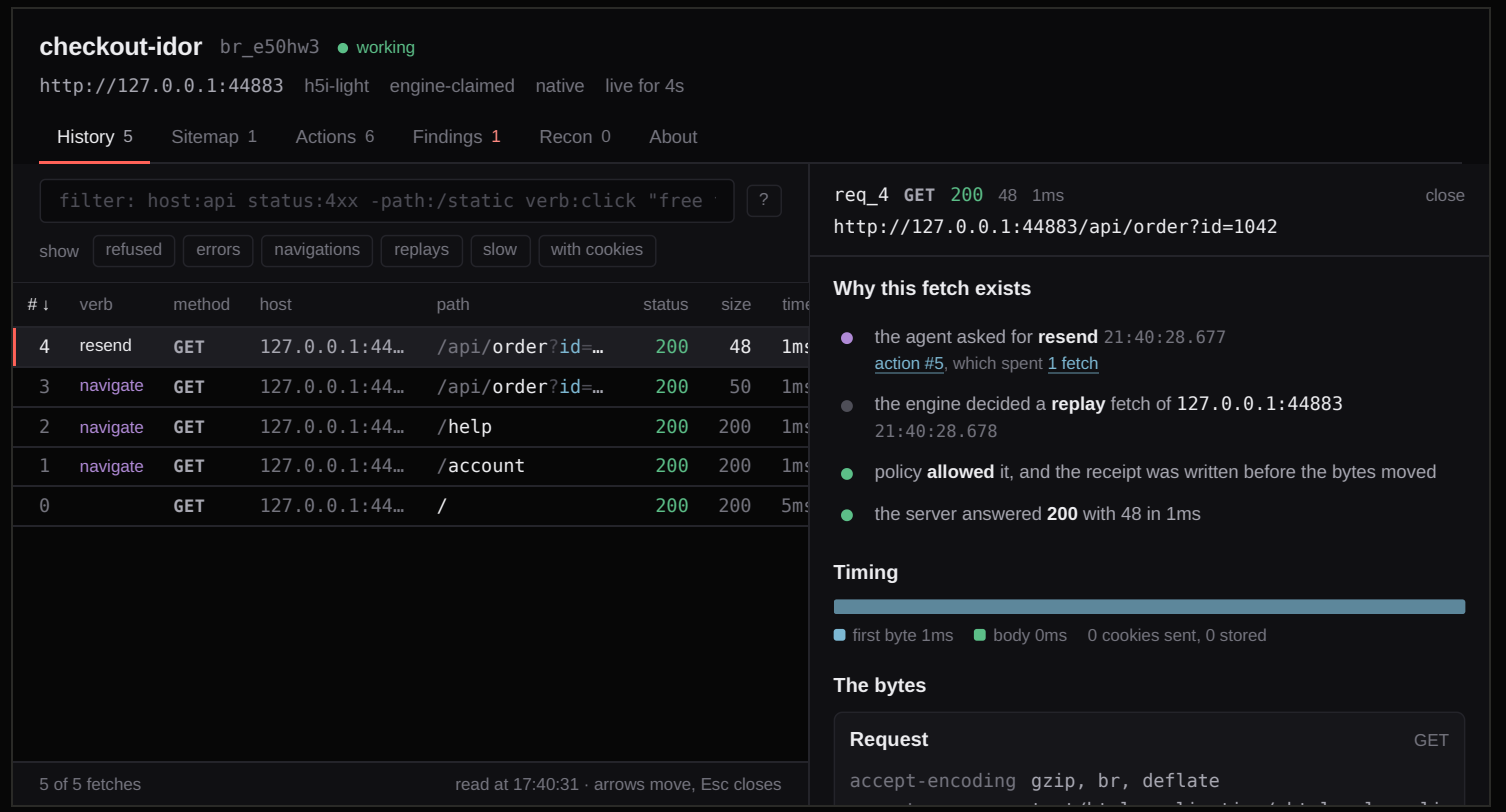
AI agent / shell

```
$ h5i browser snapshot
  @e7 Order 1041

$ h5i websec replay req_3 \
    --set query.id=1042
  → req_4 · HTTP 200

$ h5i websec diff res_3 res_4
  owner: Alice → Bob
```

↔



Prove correctness with Lean 4

Rust / kernel/src/lib.rs

```rust
fn delete(user: u64, s: &Snapshot,
          id: u64) -> Outcome {
    match find_post(&s.posts, id) {
        None => Err(Error::NotFound),
        Some(p) => {
            if p.author == user
                || is_moderator(&s.moderators, user) {
                Ok((one(Write::DelPost(id)),
                    Reply::Done))
            } else {
                Err(Error::Forbidden)
            }
        }
    }
}
```

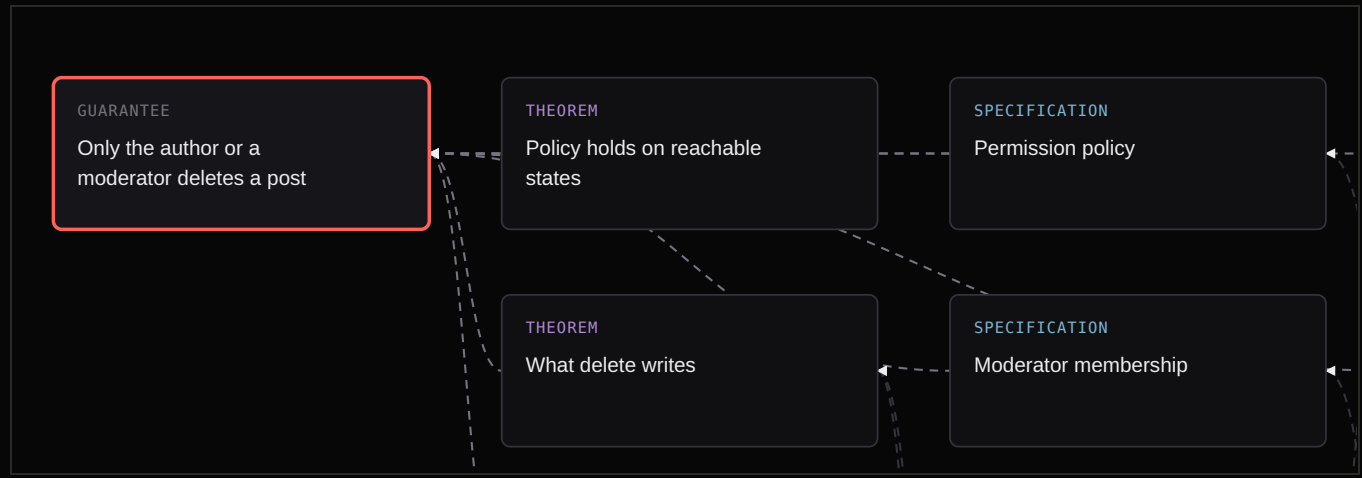
Charon
Aeneas

→

Lean 4 / Theorems.lean · excerpt

```
theorem authorized (a : Principal) (s : Snapshot)
    (c : Command) ws r
    (hinv : Inv (Snapshot.toSt s))
    (h : transition a s c = .ok (.Ok (ws, r))) :
    ∀ w ∈ ws.val, allowed (Snapshot.toSt s) a.user.val w := by -- proof body omitted
```

h5i ui / theorem-spec dependency graph · excerpt



Proofs apply to stated properties under documented assumptions

Red-Teaming for Every Team
Formal Verification for Every Team

AI-LED RED-TEAMING

Find Vulnerabilities

FORMAL VERIFICATION

Prove Correctness

Software security

07 / EARLY TRACTION

Open-source traction

1.6k+

RELEASE DOWNLOADS

700+

GITHUB STARS

Maintainer-reported · October 11, 2026 Repository ↗ Releases ↗

Find bugs. Prove correctness. Build trust.

NEXT

- Work with design partners on real applications
- Improve the workflow
- Build toward the first paid pilots

Start a conversation ↗

Explore h5i ↗